# DEVELOPMENT OF A WEB-BASED INTEGRATED ACCOUNTING SYSTEM WITH DIGITAL FINANCIAL SERVICES FOR TUP VISAYAS

**A Project in Software Engineering**

Presented to the Faculty of the

**Computer Engineering Technology Department**

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# INTRODUCTION

## Background of the Study

In today's digital age, the adoption of web-based systems has significantly transformed how institutions manage information, deliver services, and interact with users. With the continuous advancement of internet technologies and online platforms, organizations are increasingly shifting from traditional manual processes to digital systems that offer efficiency, accessibility, and convenience. Educational institutions, in particular, are embracing digital transformation to streamline administrative operations and improve service delivery for students and staff. In short, digital transformation in accounting systems enhances operational efficiency and supports better decision-making through real-time data processing and accessibility (Alsharari, 2021).

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Despite these advancements, many institutions still rely on conventional accounting processes that involve paper-based documentation and face-to-face transactions. These traditional methods often result in inefficiencies such as long queues, delays in processing, and increased risk of human error. In addition, the lack of an online platform limits accessibility, requiring students to be physically present within the campus to complete financial transactions. This becomes particularly inconvenient for students who reside far from the university, as it consumes time, effort, and additional expenses. The absence of a centralized and digital accounting system also leads to fragmented information, making it difficult to track transactions, verify records, and maintain transparency. In short ineffective accounting information handling reduces organizational efficiency and negatively affects service delivery (Soudani, 2019).

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In the context of Technological University of the Philippines Visayas (TUP Visayas), the current accounting process is primarily conducted through paper-based and in-person transactions. Students are required to visit the accounting office for activities such as payment of school fees, application for summer classes, and purchasing of school-related materials. This approach not only limits operational efficiency but also restricts the accessibility of services, especially in situations where remote transactions would be more practical and necessary. Furthermore, the lack of an integrated system results in limited availability of information, reducing the ability of students to conveniently access accounting-related services and updates.

Another concern is the absence of digital financial services that allow secure and convenient online transactions. With the increasing use of digital payment platforms, users now expect systems that support online payments, automated processing, and instant transaction records. Without such capabilities, institutions may struggle to keep up with modern standards of service delivery. The integration of digital financial technologies improves transaction efficiency, reduces processing time, and enhances user satisfaction in service-oriented systems. Additionally, the lack of a centralized system for managing financial data may lead to inconsistencies in record-keeping, delays in verification, and difficulty in generating accurate reports (Nguyen et al., 2020).

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Given these challenges, there is a clear need for a system that enhances efficiency, improves accessibility, and supports digital financial transactions within the university. To address these concerns, this study proposes the Development of a Web-Based

Integrated Accounting System with Digital Financial Services for TUP Visayas. The proposed system aims to replace traditional paper-based processes with a centralized and automated platform that enables students to perform financial transactions online. It will provide features such as online payment processing, application services, transaction history tracking, and access to relevant accounting information through a user-friendly web interface.

By integrating digital financial services and information management into a single platform, the system seeks to streamline accounting operations, reduce processing time, and improve overall user experience. It also aims to promote a paperless environment, enhance data accuracy, and provide a reliable and secure method for handling financial transactions. Ultimately, this study contributes to the digital transformation of accounting services in TUP Visayas by offering a modern solution that aligns with current technological trends and user expectations.

## Statement of the Problem

Despite the continuous advancement of digital technologies in educational institutions, the accounting processes at Technological University of the Philippines Visayas (TUP Visayas) remain limited in terms of efficiency, accessibility, and service delivery due to the absence of a fully integrated digital system. The reliance on conventional transaction methods restricts the ability of students and staff to conveniently access financial services and manage accounting-related activities in a timely and organized manner. As a result, there is a need to examine how modern web-based solutions can be utilized to enhance the overall functionality and effectiveness of accounting operations within the university.

This study therefore aims to determine how a web-based integrated accounting system with digital financial services can be developed to address the existing limitations in transaction processing, information management, and service accessibility in TUP Visayas. Specifically, it seeks to identify how such a system can transform current processes into a centralized digital platform that enables efficient handling of financial transactions, including payments, service applications, and product purchases. It also aims to determine how the system can provide accurate and organized recording, tracking, and retrieval of financial data while ensuring reliability in system operations.

Furthermore, the study intends to examine how the proposed system can enhance transaction efficiency by reducing processing time and improving user interaction through a structured and user-friendly interface. It also explores how security measures can be incorporated to ensure the protection of user information and the integrity of digital financial transactions. Lastly, the study aims to evaluate the usability and acceptability of the developed system in terms of system usefulness, information quality, interface quality, and user satisfaction to ensure that it effectively meets the needs of its intended users.

## Objectives of the Study

### General Objective

This study aims to develop a Web-Based Integrated Accounting System with Digital Financial Services for TUP Visayas.

Specifically aims to:

1. Design a web-based integrated accounting system that provides a centralized platform for managing financial transactions and accounting services in TUP Visayas.
2. Develop the system using appropriate web technologies to support digital transactions, user authentication, and data management.
3. Test the system to ensure the accuracy, reliability, and performance of its features and transaction processes.
4. Evaluate the system using the Post-Study System Usability Questionnaire (PSSUQ) in terms of system usefulness, information quality, interface quality, and user satisfaction.

**Scope and Limitations of the Study**

This study focuses on the development of a web-based integrated accounting system with digital financial services for the Technological University of the Philippines Visayas (TUP Visayas). The system is intended to provide a centralized platform that enables students to perform financial transactions online, including payment of school fees, application for summer classes, and purchasing of school-related materials such as uniforms and identification accessories. It also allows users to view transaction history and access relevant accounting information through a web interface. The system supports user authentication, ensuring proper management of transactions and records. It is designed to be accessible through up-to-date web browsers. The evaluation of the system will be conducted using the Post-Study System Usability Questionnaire (PSSUQ), with selected respondents composed of students, IT specialists, and accounting staff to assess usability and overall system performance.

The study is limited to the implementation of accounting-related services within TUP Visayas and does not cover integration with external financial institutions beyond basic digital payment processing. The system requires a stable internet connection and is dependent on the availability and reliability of third-party payment services. Additionally, the system does not include advanced financial analytics, auditing tools, or real-time financial forecasting features. The accuracy of the system depends on the correctness of the data entered by users and administrators. Furthermore, the evaluation results are based only on the selected respondents and may not fully represent the perspectives of all possible users.

## Significance of the Study

This study is beneficial to the following:

**Students.** Students will benefit from the proposed system by having access to a centralized web-based platform that allows them to perform financial transactions conveniently without the need for physical presence. The system provides an accessible means for paying school fees, applying for academic services, and purchasing school-related materials, thereby reducing time, effort, and unnecessary delays.

**Accounting Staff.** Accounting staff will benefit from the system through improved efficiency in managing financial transactions and records. The automation of processes such as verification, recording, and reporting reduces manual workload, minimizes errors, and enables faster and more organized handling of accounting operations.

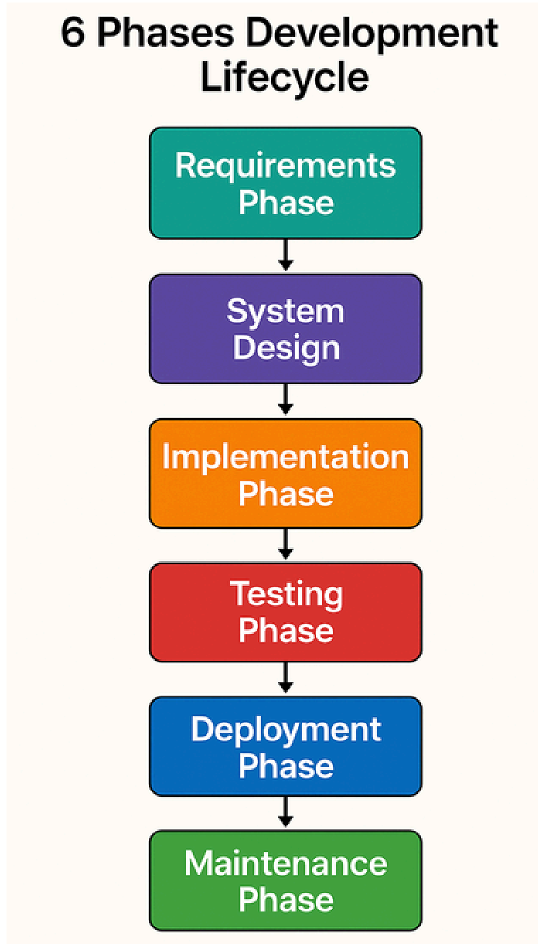
**University Administration.** The university administration will benefit from the system by gaining access to a centralized and reliable source of financial data that supports monitoring and decision-making. The availability of organized reports and real-time information enhances transparency, improves operational control, and contributes to more effective management of university resources.

**Future Researchers.** Future researchers will benefit from this study as it provides a reference for the design and development of web-based accounting systems with digital financial services. The study may serve as a foundation for further improvements, enhancements, or adaptations of similar systems in other educational institutions or organizations.

# METHODOLOGY

In the development of the proposed system, a structured and systematic approach is essential to ensure that the objectives of the study are achieved effectively and efficiently. This study adopts the Waterfall Model as the software engineering paradigm for the design and development of the Web-Based Integrated Accounting System with Digital Financial Services for TUP Visayas. The Waterfall Model is a linear and sequential approach in software development where each phase is completed before proceeding to the next. This methodology is appropriate for the study since the system requirements are clearly defined, and the development process follows a logical progression from planning to implementation.

The Waterfall Model provides a clear framework that guides the development process through a series of well-defined phases, ensuring that each stage is properly documented and evaluated. This approach allows the researchers to maintain control over the development process, minimize errors, and ensure that the system meets the specified requirements. The phases involved in this methodology include Requirements Analysis, System Design, Implementation, Testing, Deployment, and Maintenance.




*https://tinyurl.com/fkn33dhh*

*Figure 1.* Waterfall Model

Figure 1 above illustrates the Waterfall Model used in the study. It presents a step-by-step process where each phase flows sequentially into the next, starting from requirements analysis and ending with system maintenance. Each phase produces specific outputs that serve as the foundation for the succeeding stage, ensuring a structured and organized development process.

**Requirements Phase**

Requirements Analysis involves the identification and analysis of the system requirements based on the current processes and user needs. In this phase, the researchers gather relevant information regarding the existing accounting system of TUP Visayas, including its limitations and challenges. The requirements are then defined in terms of system functionalities such as online transactions, application processing, product purchasing, and transaction tracking. This phase ensures that the system to be developed addresses the actual needs of its intended users.

**Design Phase**

System Design focuses on translating the identified requirements into a structured system architecture. This includes the design of data flow, database structure, and user interface. Tools such as Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD) are utilized to represent how data moves within the system and how it is organized. The design phase serves as the blueprint of the system, guiding the development process and ensuring that all components are properly planned before implementation.

**Implementation Phase**

Implementation involves the actual development of the system based on the approved design. In this phase, the system is developed using web technologies such as HTML, CSS, and JavaScript for the frontend, PHP for the backend, and MySQL for database management. The functionalities of the system, including user authentication, transaction processing, and data storage, are coded and integrated to form a working application. This phase transforms the system design into a functional software solution.

**Testing Phase**

Testing is conducted to ensure that the system operates correctly and meets the defined requirements. Various tests are performed to verify the accuracy of transactions, reliability of system features, and responsiveness of the interface. Errors and issues identified during testing are addressed and corrected to improve system performance. This phase is essential in ensuring the quality and dependability of the developed system.

**Deployment Phase**

Deployment involves the installation and implementation of the system in a usable environment. The system is prepared for actual use by configuring the necessary software and hardware requirements. Users are introduced to the system, and initial usage is observed to ensure that it functions as intended. This phase marks the transition of the system from development to actual operation.

**Maintenance Phase**

Maintenance focuses on the continuous improvement and support of the system after deployment. This includes fixing errors that may arise during actual use, updating system features, and ensuring that the system remains functional and relevant over time. Maintenance ensures the long-term usability and effectiveness of the system in meeting the needs of its users.

## DATA GATHERING PROCEDURES AND OUTPUTS

In the development of the proposed system, data gathering is essential in identifying the requirements needed for the system. This phase focuses on collecting relevant information about existing procedures, user needs, and system expectations to ensure that the proposed solution is aligned with actual operational conditions.

To obtain accurate data, the study utilized three methods: interview, observation, and document analysis. These methods provided a clear understanding of the current accounting process and served as the basis for defining system requirements.

The interview method was conducted with individuals involved in the accounting process, primarily accounting cashiers and staffs of TUP Visayas responsible for handling financial transactions. Through this method, the researchers were able to gather detailed information regarding the current workflow, common issues encountered during transactions, and the specific needs of users. The interview also provided insights into how transactions are verified, recorded, and managed, which served as a basis for identifying system requirements and necessary features.

The observation method was used to directly examine how accounting transactions are performed within the university. This includes observing the step-by-step process of handling payments, processing applications, and managing records in a paper-based and face-to-face environment. This helps to identify inefficiencies such as delays, repetitive tasks, and manual errors. This method ensured that the proposed system reflects the actual workflow and addresses real operational challenges.

The document analysis method involved reviewing existing forms and records used in the current accounting system, such as payment slips, transaction records, and other related documents. This allowed the researchers to identify the types of data being collected, the format of records, and the essential information required in transactions. The analysis of these documents provided a clear basis for designing the database structure and ensuring that all necessary data fields are included in the proposed system.

Based on the gathered data, the system requirements were identified and categorized into functional and non-functional requirements. The functional requirements include processing of online transactions such as payment of fees, application for services, and purchasing of school materials; user account management; generation of transaction records and receipts; and access to transaction history and accounting information. The non-functional requirements include system security through authentication and role-based access, usability through a clear interface, efficient system performance, and web-based accessibility. The requirements are also defined according to user roles. Students are provided with access to perform transactions and view records, while accounting staff are given tools to manage transactions and maintain financial data. Overall, the data gathered in this phase established the foundation for the system's analysis and design, ensuring that the proposed solution is based on actual requirements.

## DOCUMENTATION OF THE CURRENT SYSTEM

The current accounting system of Technological University of the Philippines Visayas (TUP Visayas) operates through a manual and in-person process where transactions are handled directly at the accounting office. Financial activities such as payments, service applications, and purchasing of school-related materials are performed through physical interaction between students and accounting staff.

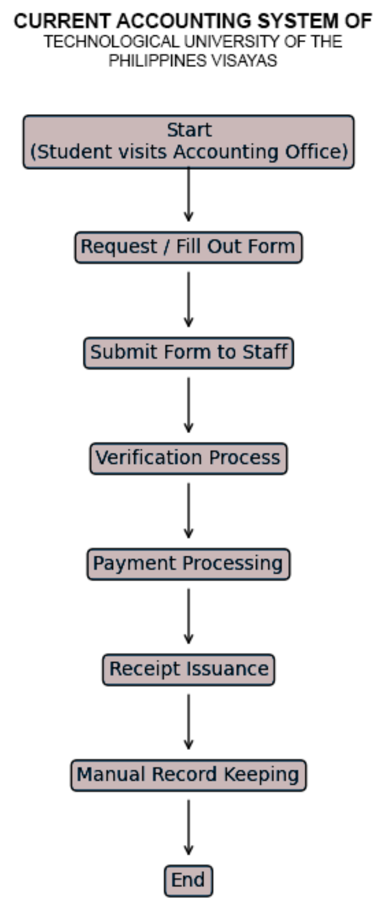



*Figure 2.* Current Accounting System of TUP Visayas

Figure 2 above presents the general flow of the current accounting system, showing how transactions are initiated, processed, and recorded through a sequence of manual steps. The process typically begins with the student initiating a request, followed by form completion, verification, payment, and record keeping.

In the process of payment of school fees, a student first visits the accounting office and requests the necessary form or payment details. The student fills out the required information manually and submits it to the cashier. The cashier then reviews and verifies the provided information before proceeding with the payment. For application of academic services, such as summer class enrollment, the student is required to secure and complete an application form. The submitted form undergoes verification to determine eligibility and availability of slots. In terms of purchasing school-related materials, the student directly coordinates with the accounting office to inquire about available items such as uniforms or identification accessories. Once the payment is completed for all types of transactions, a receipt is issued, and the transaction is recorded in the system through manual entry or documentation.

Based on the documented processes, several operational concerns can be observed. The reliance on manual input and physical records increases the possibility of errors and inconsistencies in data handling. Transaction processing depends on step-by-step human interaction, which may affect processing time during peak periods. Access to services is limited to on-site transactions, and retrieving past records may require additional time and effort. Overall, the current system follows a structured but manual workflow that relies on physical presence, document handling, and sequential processing of transactions. These characteristics highlight the need for a more streamlined and accessible approach to managing accounting services, which will be addressed in the proposed system.